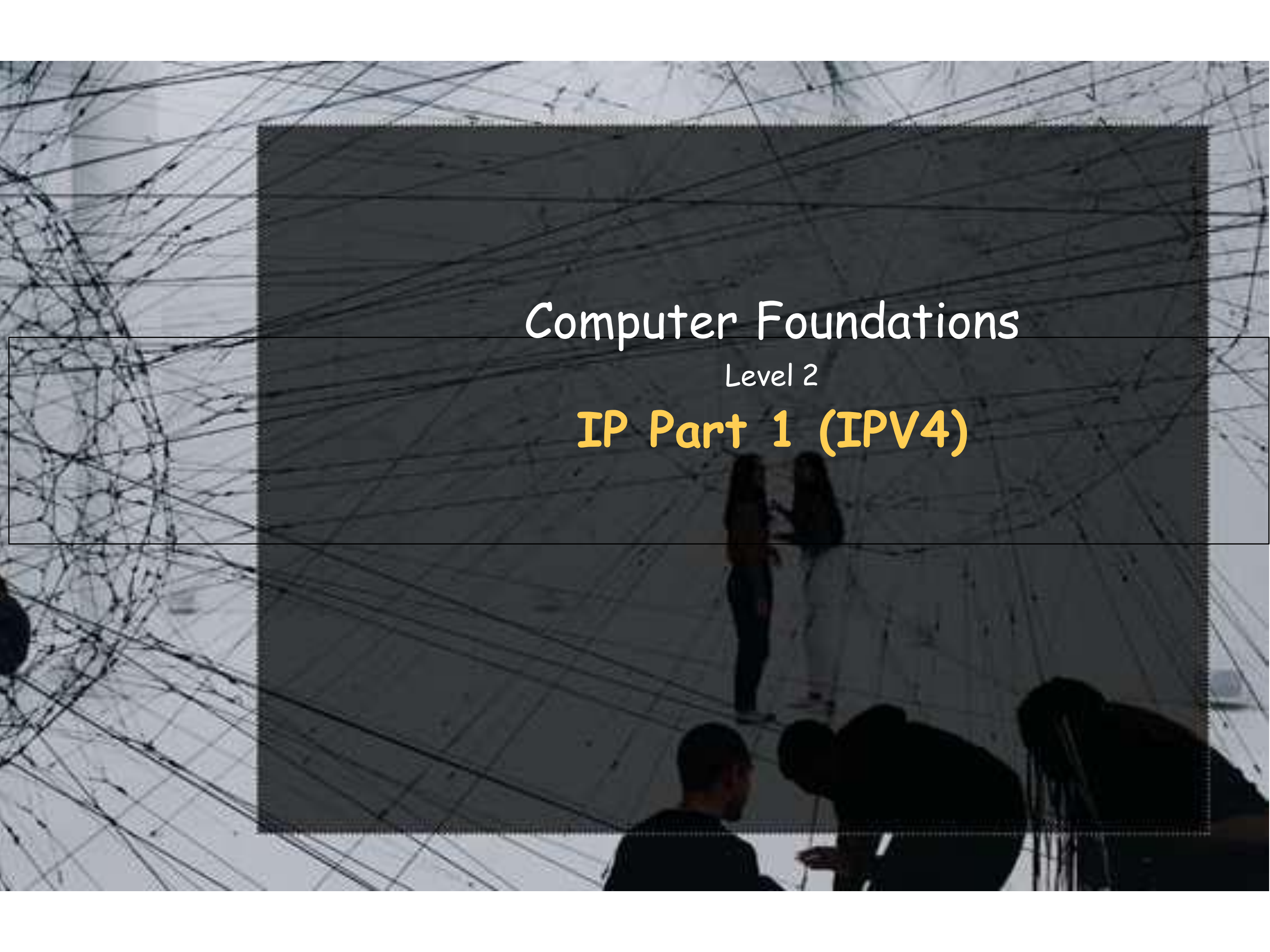




Computer Foundations Level 2



Computer Foundations

Level 2

IP Part 1 (IPV4)

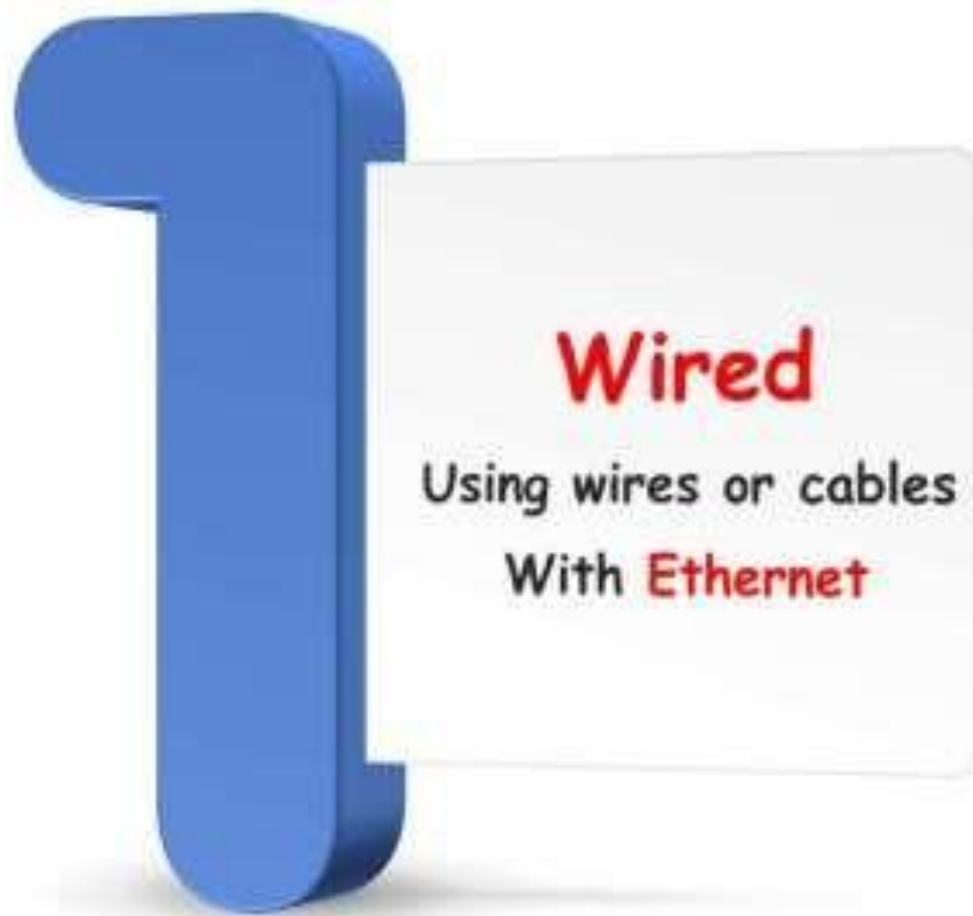


Computer Foundations

Level 2

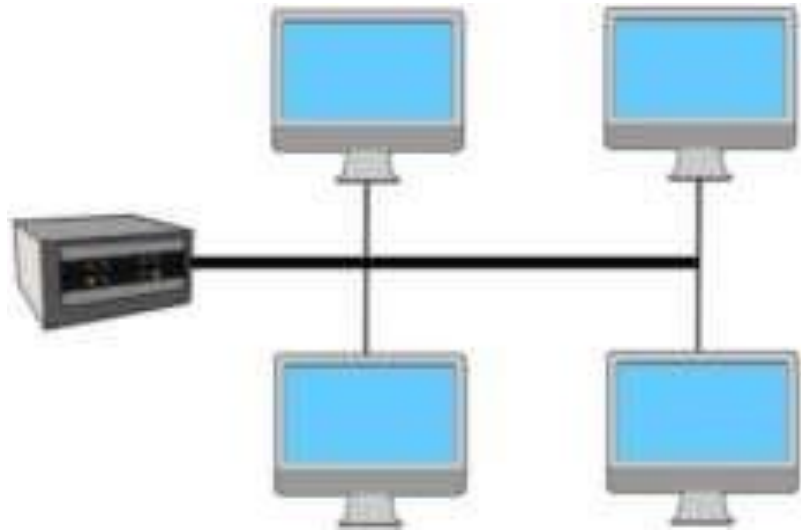
Communicating on LAN

Connection Types



Communication Ways

Switch



Ethernet

Rules that computers follow for communicating on a wired network



Router

Wireless

Rules that computers follow for communicating through a radio signals

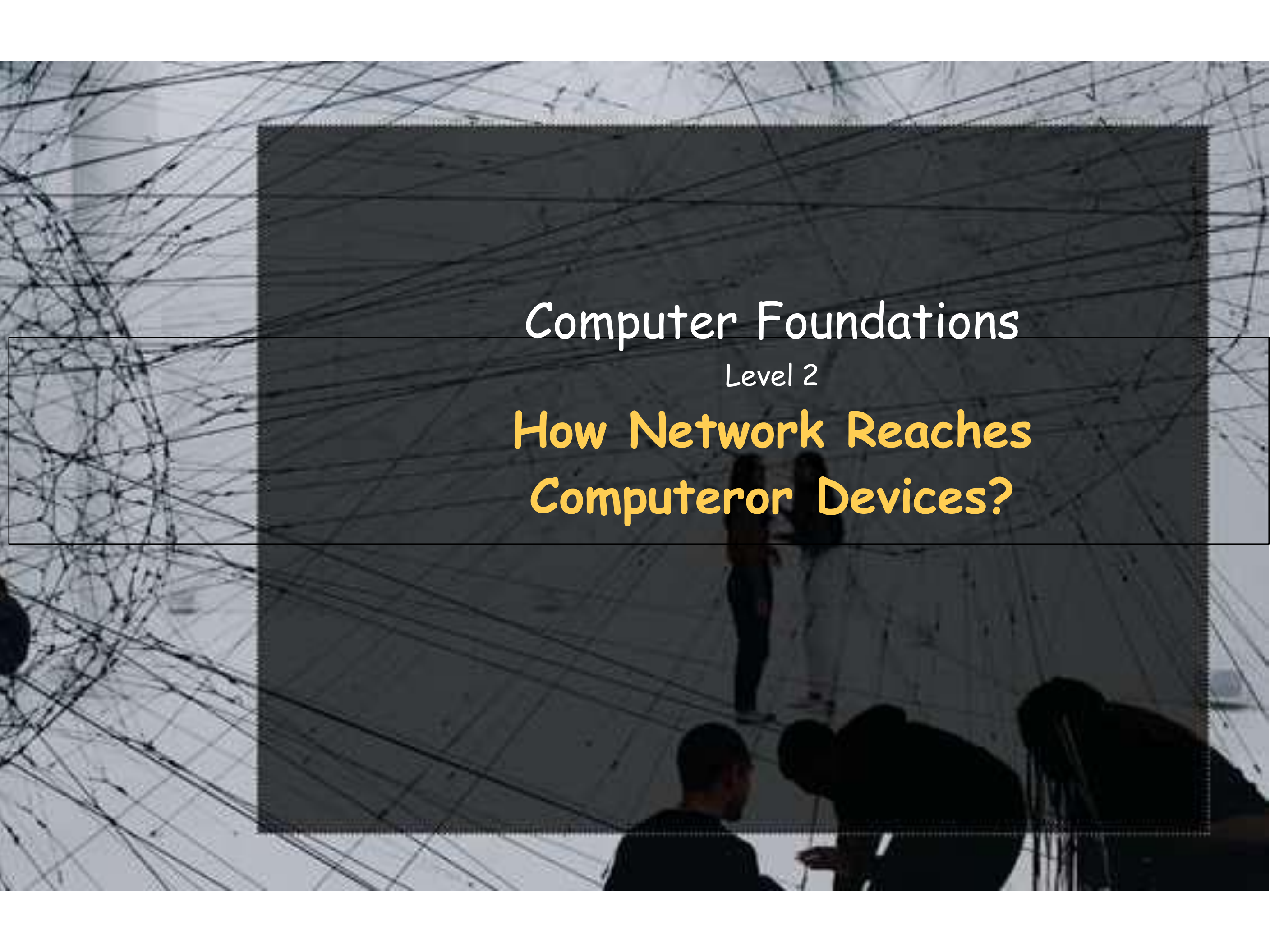
Wi-Fi is a popular way to connect wirelessly to a network.

Wi-Fi: Wireless Fidelity

Those Rules are called:

Protocol

Protocol: is a standard set of rules for how devices communicate through wires or wireless



Computer Foundations

Level 2

How Network Reaches Computer or Devices?

How Computers Know the address of each others?



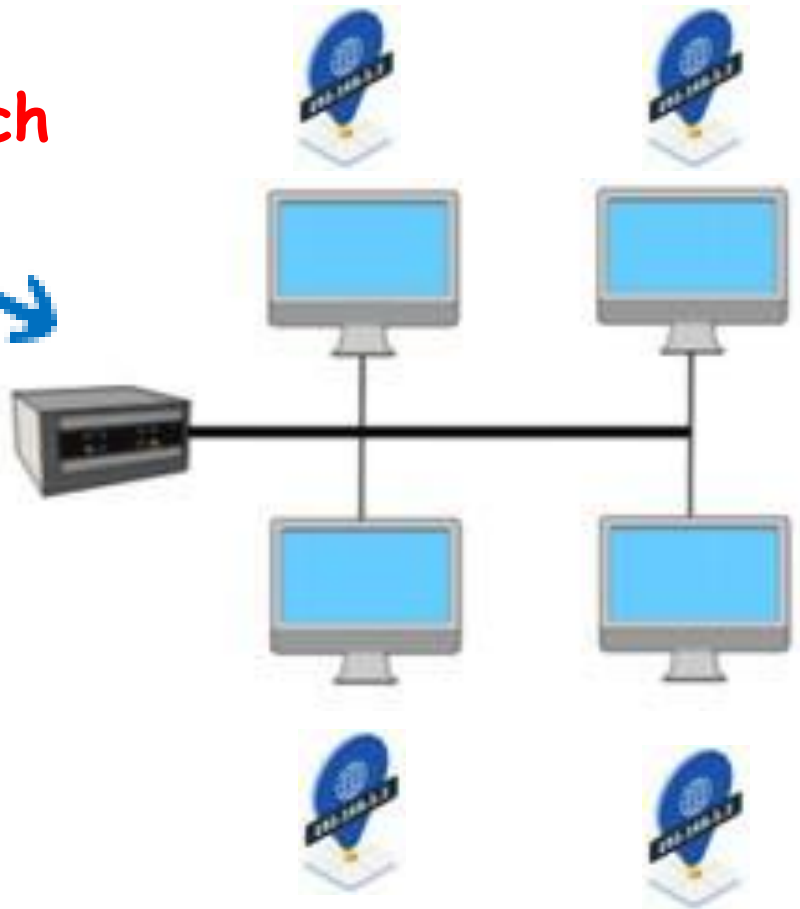
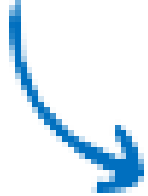
Packet2

Packet1



How to locate certain device?

Switch



What is IP Address?

- Every computer on an Ethernet network is given an IP address.
- An IP address, like your home's street address, identifies network machines.
- Because each computer has its own IP address, it aids traffic flow between them.
- (**IP address**) is a numerical identifier that is connected with a certain computer or computer network.
- When computers are connected to the internet or Network, the IP address allows them to send and receive data.



Internet

[Set a data limit to help control data usage on this network](#)

IP settings

IP assignment:

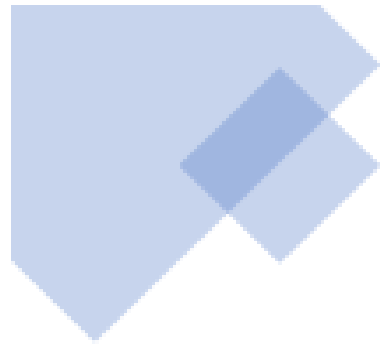
Automatic (DHCP)

[Edit](#)

Properties

SSID:	Internet
Protocol:	Wi-Fi 4 (802.11n)
Security type:	WPA2-Personal
Network band:	2.4 GHz
Network channel:	11
Link speed (Receive/Transmit):	1300/144 (Mbps)
Link-local IPv6 address:	fe80::708e:8e15:8a48:57f4%15
IPv4 address:	192.168.1.144
IPv4 DNS servers:	192.168.1.1
Manufacturer:	Broadcom
Description:	Broadcom 802.11ac Network Adapter
Driver version:	101.10.603.0
Physical address (MAC):	88-66-5A-03-A3-AB

[Copy](#)



IPv4 Address Format (Dotted Decimal Notation)

123.89.46.72

First Octet Second Octet Third Octet Fourth Octet

01111011.01011001.00101110.01001000

1 Byte=8 Bits

4 Bytes =32 Bits



What is IP Stands for?

IP: is **I**nternet **P**rotocol.

- The acronym IP stands for "**I**nternet **P**rotocol," which is a collection of rules that regulate the format of data transferred over the internet or a local network.

Thank you 😊